

# Automated Performance and Footprint Benchmark Report

- Generated At: 2026-08-31 14:55:48
- Application Package: app.phial.habits
- Target Device: sdk\_gphone64\_arm64 (Android 16, 1080x2400, Serial: emulator-5554)
- Report Identifier / Tag: emulator
- Interaction Driver: Maestro (benchmark\_suite.yaml)

## Executive Summary

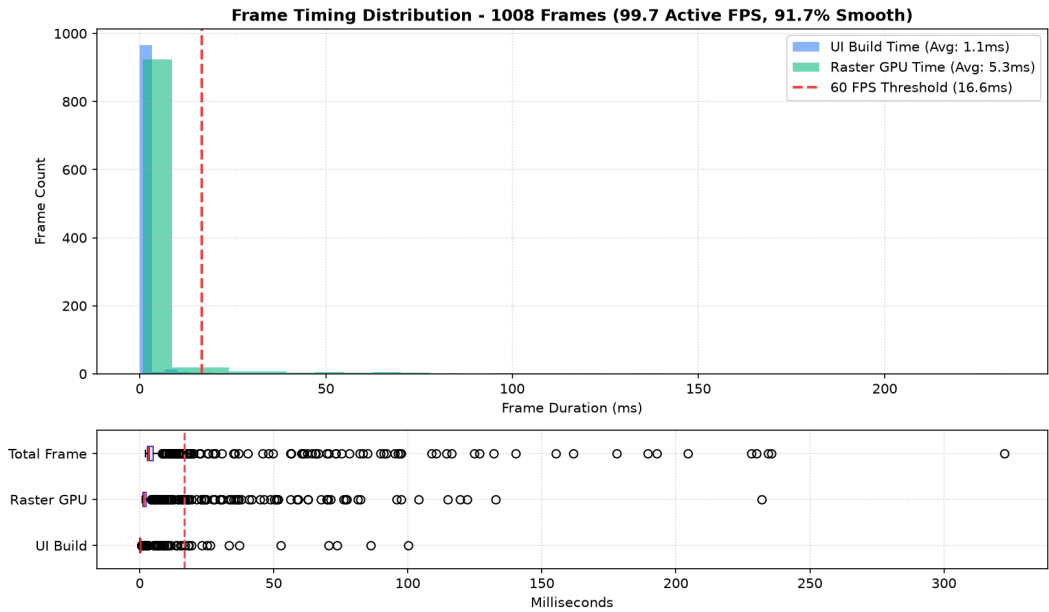
| Metric                     | Baseline | Peak     | Resting  | Delta (Post-GC) | Health Status |
|----------------------------|----------|----------|----------|-----------------|---------------|
| Total System RAM (PSS)     | 167.8 MB | 186.8 MB | 166.8 MB | -1.0 MB         | STABLE        |
| Native Heap (Impeller/C++) | 37.9 MB  | -        | 31.5 MB  | -6.4 MB         | OPTIMAL       |
| Dart VM Heap               | 21.9 MB  | -        | 21.0 MB  | -0.9 MB         | OPTIMAL       |

## CPU Utilization

- Average Active CPU: 14.2%
- Peak CPU Spike: 20.0%

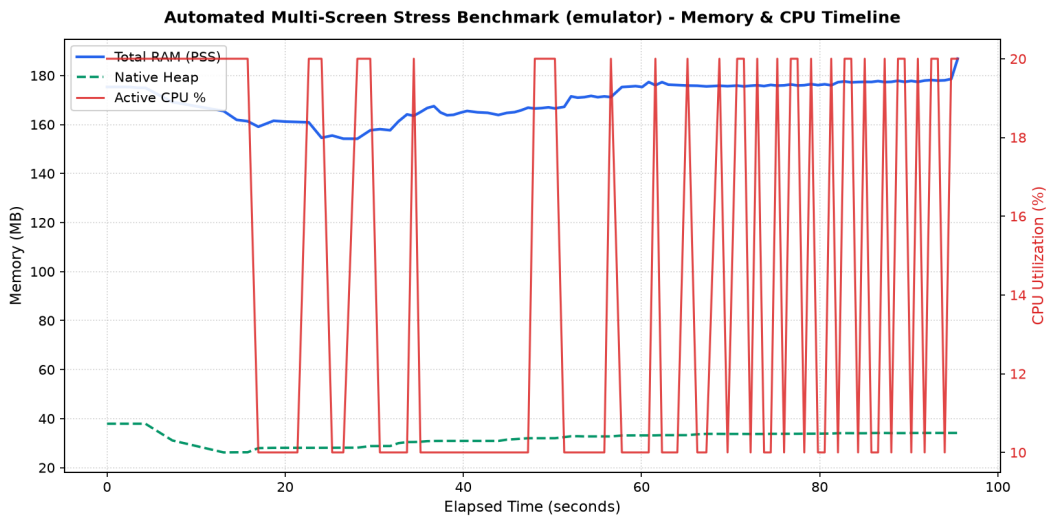
## Frame Rate and Rendering Performance

| Rendering Metric            | Measurement             | Target / Budget  | Status    |
|-----------------------------|-------------------------|------------------|-----------|
| Total Frames Rendered       | 1,008                   | -                | COMPLETED |
| Active Motion Render Rate   | 99.7 FPS                | 60.0 / 120.0 FPS | OPTIMAL   |
| Frame Smoothness / Fluidity | 91.7%                   | > 95.0%          | WARN      |
| Janky Frames (>16.6ms)      | 84 (8.3%)               | < 5.0%           | WARN      |
| UI Thread Build Time (Avg)  | 1.1 ms (95th: 2.2ms)    | < 8.0 ms         | OPTIMAL   |
| Raster GPU Time (Avg)       | 5.3 ms (95th: 22.9ms)   | < 8.0 ms         | OPTIMAL   |
| Total Frame Time (Avg)      | 10.0 ms (99th: 155.3ms) | < 16.6 ms        | OPTIMAL   |



Frame Timing Distribution

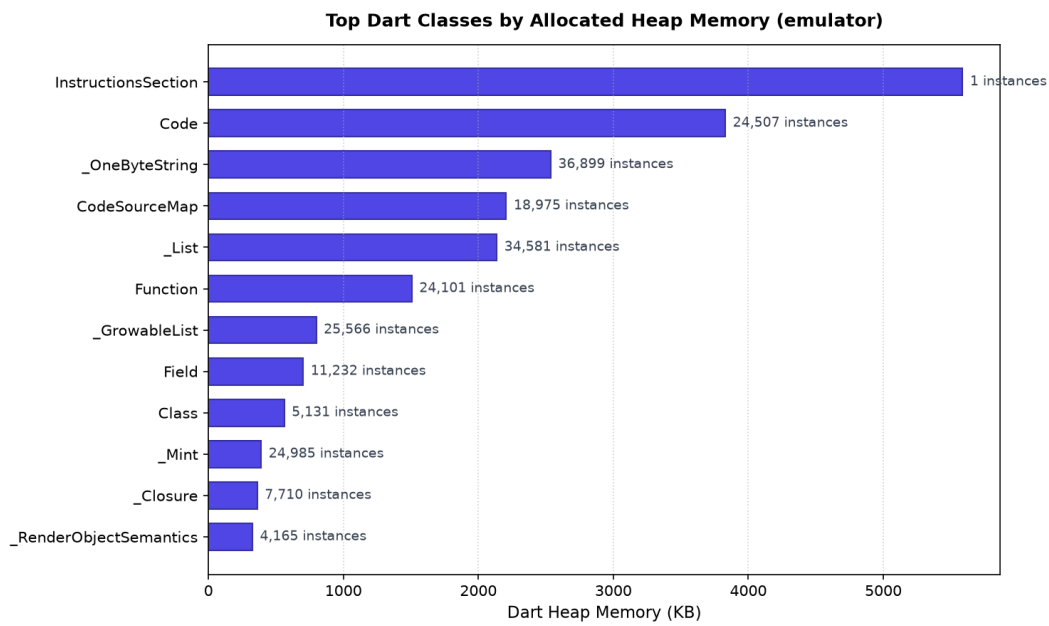
## Performance Timeline



Memory and CPU Timeline

## Dart Heap Class Allocations

| Class Name             | Active Instances | Memory Allocated (KB) |
|------------------------|------------------|-----------------------|
| InstructionsSection    | 1                | 5587.2 KB             |
| Code                   | 24,507           | 3829.2 KB             |
| _OneByteString         | 36,899           | 2535.6 KB             |
| CodeSourceMap          | 18,975           | 2206.4 KB             |
| _List                  | 34,581           | 2136.1 KB             |
| Function               | 24,101           | 1506.3 KB             |
| _GrowableList          | 25,566           | 798.9 KB              |
| Field                  | 11,232           | 702.0 KB              |
| Class                  | 5,131            | 561.2 KB              |
| _Mint                  | 24,985           | 390.4 KB              |
| _Closure               | 7,710            | 361.4 KB              |
| _RenderObjectSemantics | 4,165            | 325.4 KB              |



Top Dart Allocations